Forecasting Lab Delays Using Workflow & Device Data

An AI-Powered Predictive Framework for Operational Risk in R&D Laboratories

Group 2 | MBDS Capstone 2026 | IE University x Roche Pharma S.A. (Lab Operations)

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1. Business Context

In pharmaceutical R&D laboratories, experiment delays arise from instrument failures, reagent shortages, and scheduling bottlenecks, compounding throughout the day and extending critical project timelines.

Roche's Lab Experience team sought to move beyond reactive management by leveraging predictive analytics to provide early warnings and enable proactive resource reallocation.

Can we classify experiments as 'at-risk' before they begin, and identify the key contributors to delay?

2. Data

350K Experiments	**350K** Telemetry Events
50 Reagent Batches	**30** Instruments
5 Experiment Types	**365** Days of Data

Data Architecture

Workflow Logs
- 350K experiments
- Booking times
- Durations
- Workload levels
- Instrument IDs

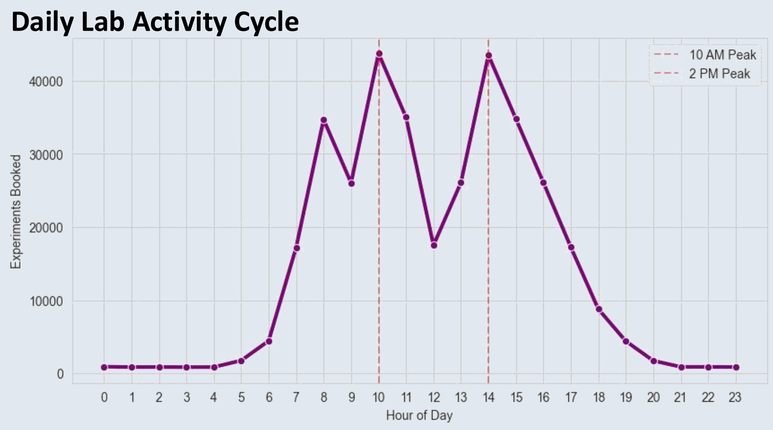
Telemetry Logs
- 350K device events
- Ambient temp
- Health scores
- Incident logs
- Timestamps

Reagent Logs
- 50 batches tracked
- Stock levels
- Out-of-stock flags
- Batch quality
- Availability

Merged Dataset | 350K Rows | 25 Features

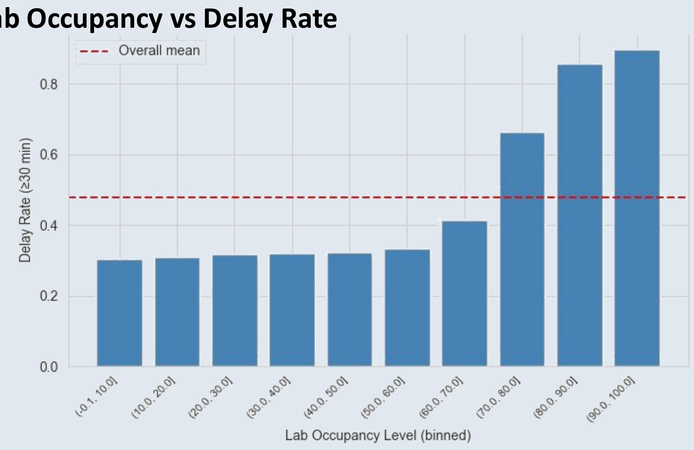
Exploratory Data Analysis

Daily Lab Activity Cycle



Bimodal activity peaks at 10 AM and 2 PM with a lunch dip, experiments during peaks face higher queue contention and delay risk.

Lab Occupancy vs Delay Rate

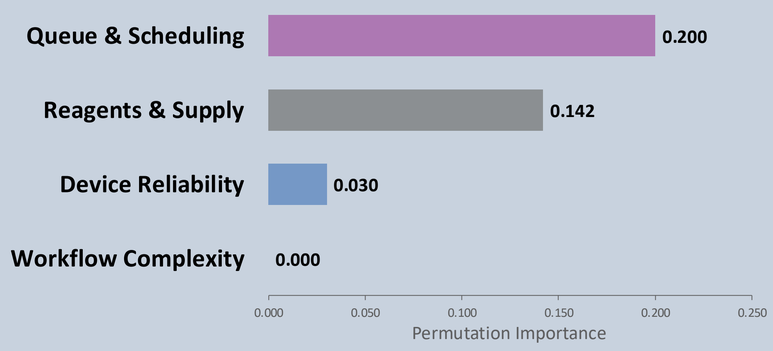


Delay risk remains flat below 60% occupancy then spikes exponentially, confirming congestion as the dominant driver of lab delays.

3. Model

Feature Importance

We engineered 25 features across four families, Queue & Scheduling (11), Device Reliability (10), Workflow Complexity (3), and Reagents & Supply (1). Permutation importance reveals which families drive the model's predictions:



Model Comparison

Three classifiers evaluated with temporal 70/15/15 split:

Model	ROC-AUC	PR-AUC	F1
Logistic Regression	0.803	0.827	0.747
Random Forest	0.809	0.831	0.742
Gradient Boosting	**0.822**	**0.844**	**0.759**

Champion Model

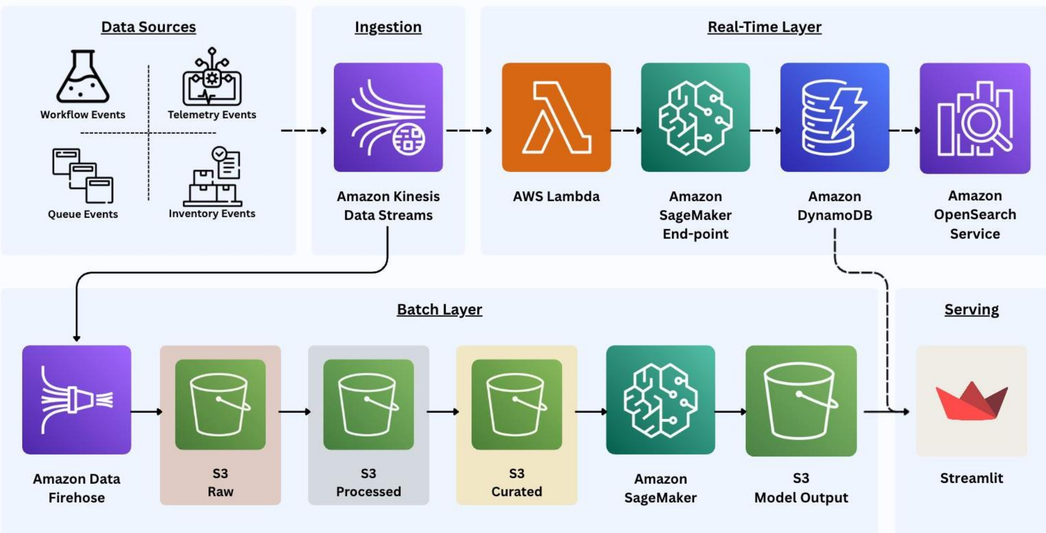
0.863	0.883	0.797
ROC-AUC	PR-AUC	Best F1

4. Pipeline & Architecture

End-to-End

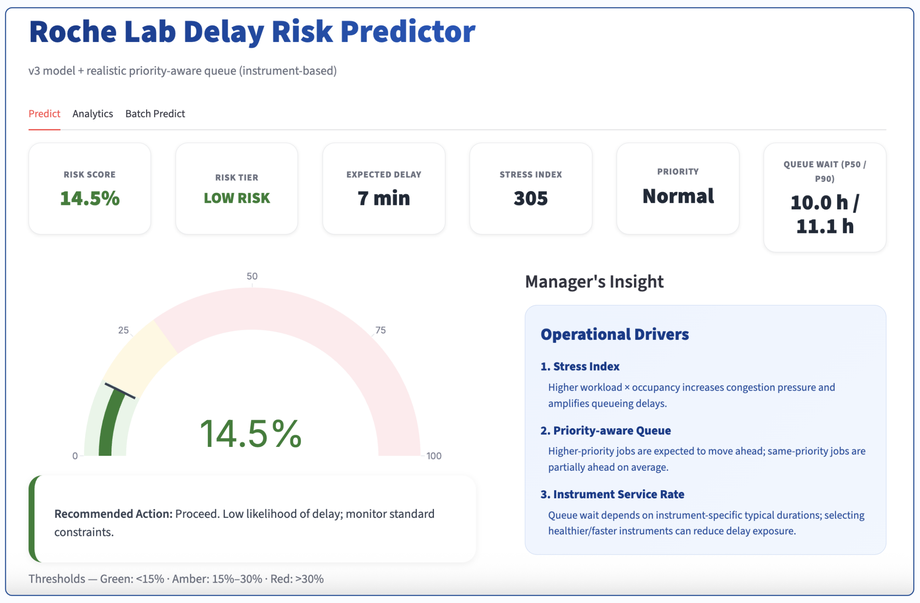
Data Collection → Feature Engineering → Gradient Boosting → Streamlit Dashboard → AWS Deployment

Production Architecture (AWS)



Unified AWS streaming architecture, which uses Kinesis ingestion, Lambda for real-time scoring, SageMaker for model serving, and a three-tier S3 data lake with enterprise governance.

Deployed Application



Interactive risk predictor enabling scientists to assess delay probability, simulate what-if scenarios, and receive AI-driven mitigation recommendations before experiments begin.

5. Key Insights

Lab Occupancy is the #1 Delay Driver.

This single variable (importance: 0.153) explains more delay variance than any other, making occupancy-aware scheduling the highest-leverage intervention.

Instruments Signal Failure Before It Happens

Cumulative usage creates measurable drift in delay probability. Rolling 30-day failure trends enable proactive maintenance before critical thresholds are reached.

Defective Reagent Batches Are Detectable

3 out of 50 batches carry 5x the failure risk. The model flags them automatically, enabling quality alerts before experiments are affected.

6. Recommendations

1. **Smart Scheduling** — Reduce peak-hour wait times by up to 30%
2. **Reagent Quality Monitoring** — Flag suspect batches before experiments begin
3. **Predictive Maintenance** — Service instruments before they fail
4. **Off-Peak Scheduling** — Spread experiments away from 10 AM and 2 PM

7. Conclusion

Our framework predicts experiment delays with 86.3% AUC, catching 84% of actual delays before they occur, enabling Roche to reduce disruptions, optimize scheduling, and flag defective reagent batches proactively.

Just two factors, **lab congestion and reagent batch quality,** account for over 90% of predictive power, meaning targeted interventions in scheduling and supply chain monitoring alone could dramatically reduce delay frequency.

The system is deployment-ready: a Streamlit dashboard delivers real-time risk scoring, what-if scenario planning, and AI-driven mitigation recommendations, backed by a scalable AWS architecture for enterprise integration.

This solution gives scientists more time for science.